

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0648

Commonwealth Chesapeake 345 kV
December 02, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0648. The Interconnection Customer proposes a 318.6 MW solar photovoltaic facility interconnecting at the Commonwealth Chesapeake 345 kV substation in the Virginia area. The request was received on July 18, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 280 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$30,127,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: W -STATION to ST BONIFACIUS 138 KV
Contingency: SOUTHTOWN to VIENNA OPERATIONS SUBSTATION 345 KV
Loading: 226.8% (Rating: 417 MVA)
DFAX: 54.7%
Pre-project loading: 185.1%
2. Facility: EQUUS FREEPORT POWER to CANNELTON HYDROELECTRIC PLANT 138 KV
Contingency: VIENNA OPERATIONS SUBSTATION to EASTON 2 345 KV
Loading: 219.7% (Rating: 259 MVA)
DFAX: 25.0%
Pre-project loading: 188.9%
3. Facility: HENNING SUBSTATION to KIWI 138 KV
Contingency: VIENNA OPERATIONS to EASTON 2 345 KV
Loading: 218.4% (Rating: 262 MVA)
DFAX: 33.1%
Pre-project loading: 178.0%
4. Facility: CANNELTON HYDROELECTRIC PLANT to HENRY ST WPS 138 KV
Contingency: VIENNA OPERATIONS to EASTON 2 SUBSTATION 345 KV line
Loading: 216.7% (Rating: 248 MVA)
DFAX: 25.0%
Pre-project loading: 184.6%
5. Facility: EQUUS FREEPORT POWER to KNIGHT STREET 138 KV
Contingency: VIENNA OPERATIONS to EASTON 2 SUBSTATION 345 KV
Loading: 210.7% (Rating: 279 MVA)
DFAX: 25.0%
Pre-project loading: 182.0%

6. Facility: SUNSET to ST BONIFACIUS 138 KV
Contingency: SOUTHTOWN to VIENNA OPERATIONS 345 KV
Loading: 209.6% (Rating: 442 MVA)
DFAX: 54.7%
Pre-project loading: 170.1%
7. Facility: KNIGHT ST to LONG LAKE 138 KV L
Contingency: VIENNA OPERATIONS to EASTON 2 345 KV Line
Loading: 206.7% (Rating: 299 MVA)
DFAX: 25.0%
Pre-project loading: 180.0%
8. Facility: HILLSBORO to PARK AVE 138 KV
Contingency: CADY to EASTON 2 345 KV
Loading: 206.2% (Rating: 296 MVA)
DFAX: 25.0%
Pre-project loading: 179.3%
9. Facility: NAVY MOLE to HOLSTON DAM 138 KV
Contingency: EASTON 2 to CADY 345 KV line
Loading: 201.5% (Rating: 390 MVA)
DFAX: 25.2%
Pre-project loading: 180.9%
10. Facility: HENRY ST WPS to QUINN 138 KV
Contingency: VIENNA OPERATIONS SUBSTATION to EASTON 2 345 KV
Loading: 196.8% (Rating: 262 MVA)
DFAX: 25.0%
Pre-project loading: 166.4%
11. Facility: YANKE to JARRATT 138 KV
Contingency: VIENNA OPERATIONS to EASTON 2 345 KV
Loading: 196.4% (Rating: 285 MVA)
DFAX: 27.2%
Pre-project loading: 166.0%
12. Facility: LINDE to RIPPLE 138 KV
Contingency: VIENNA OPERATIONS SUBSTATION to EASTON 2 345 KV
Loading: 190.2% (Rating: 337 MVA)
DFAX: 25.0%
Pre-project loading: 166.6%
13. Facility: LONG LAKE to MCMINNVILLE 138 KV
Contingency: VIENNA OPERATIONS to EASTON 2 345 KV
Loading: 190.1% (Rating: 329 MVA)
DFAX: 25.0%
Pre-project loading: 165.8%
14. Facility: PRUNEDALE to VALLEYVIEW 138 KV
Contingency: SOUTHTOWN to COMMONWEALTH CHESAPEAKE 345 KV
Loading: 187.9% (Rating: 235 MVA)
DFAX: 20.6%
Pre-project loading: 160.0%
15. Facility: PARK AVE to YANKE 138 KV Line
Contingency: VIENNA OPERATIONS to EASTON 2 345 KV line
Loading: 187.5% (Rating: 295 MVA)
DFAX: 27.2%
Pre-project loading: 158.1%
16. Facility: LINDE to MCMINNVILLE 138 KV L
Contingency: VIENNA OPERATIONS to EASTON 2 345 KV

- Loading: 186.8% (Rating: 339 MVA)
DFAX: 25.0%
Pre-project loading: 163.3%
17. Facility: EASTON 2 to NAVY MOLE 138 KV
Contingency: EASTON 2 to CADY 345 KV
Loading: 184.0% (Rating: 412 MVA)
DFAX: 44.0%
Pre-project loading: 149.9%
18. Facility: EASTON 2 345/138 kV Transformer Bank
Contingency: EASTON 2 to CADY 345 KV
Loading: 182.0% (Rating: 416 MVA)
DFAX: 44.0%
Pre-project loading: 148.3%
19. Facility: WATTSVILLE to MENOMONIE 138 KV
Contingency: SUTHTOWN to COMMONWEALTH CHESAPEAKE SUBSTATION 345 KV Line
Loading: 181.8% (Rating: 302 MVA)
DFAX: 24.1%
Pre-project loading: 156.4%
20. Facility: GRASONVILLE to MACON 138 KV
Contingency: CADY to EASTON 2 345 KV
Loading: 179.7% (Rating: 197 MVA)
DFAX: 12.6%
Pre-project loading: 159.2%
21. Facility: VIENNA OPERATIONS 138/1 kV Autotransformer
Contingency: VIENNA OPERATIONS to EASTON 2 345 KV
Loading: 177.0% (Rating: 452 MVA)
DFAX: 50.0%
Pre-project loading: 141.8%
22. Facility: MENOMONIE to COMMONWEALTH CHESAPEAKE 138 KV
Contingency: SOUTHTOWN to COMMONWEALTH CHESAPEAKE 345 KV
Loading: 176.1% (Rating: 321 MVA)
DFAX: 24.1%
Pre-project loading: 152.2%
23. Facility: ARCOGEN to CHINCOTEAGUE 138 KV
Contingency: SOUTHTOWN to COMMONWEALTH CHESAPEAKE 345 KV
Loading: 174.5% (Rating: 197 MVA)
DFAX: 16.1%
Pre-project loading: 148.4%
24. Facility: SOUTHTOWN SUBSTATION to VIENNA OPERATIONS 345 KV
Contingency: OCEAN CITY to CANDELARIA ARENAS 1718 138 KV
Loading: 171.4% (Rating: 1,020 MVA)
DFAX: 88.8%
Pre-project loading: 143.7%
25. Facility: HUNTINGTON FALLS to HUBBELL 138 KV
Contingency: SOUTHTOWN to COMMONWEALTH CHESAPEAKE 345 KV
Loading: 169.1% (Rating: 186 MVA)
DFAX: 12.4%
Pre-project loading: 147.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W -STATION to ST BONIFACIUS 138 KV	Rebuild line with higher-capacity conductor	\$7,843,000	14% (\$1,064,000)
CANNELTON HYDROELECTRIC PLANT to EQUUS FREEPORT POWER 138 KV	Rebuild line with higher-capacity conductor	\$2,612,000	13% (\$337,000)
HENNING to KIWI SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$3,105,000	14% (\$422,000)
CANNELTON HYDROELECTRIC PLANT to HENRY ST WPS SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$6,354,000	13% (\$821,000)
EQUUS FREEPORT POWER to KNIGHT STREET 138 KV	Rebuild line with higher-capacity conductor	\$3,745,000	13% (\$484,000)
SUNSET to ST BONIFACIUS 138 KV	Rebuild line with higher-capacity conductor	\$5,389,000	14% (\$731,000)
KNIGHT ST to LONG LAKE 138 KV L	Rebuild line with higher-capacity conductor	\$16,109,000	13% (\$2,080,000)
HILLSBORO to PARK AVE 138 KV	Rebuild line with higher-capacity conductor	\$5,909,000	13% (\$762,000)
NAVY MOLE to HOLSTON DAM 138 KV	Rebuild line with higher-capacity conductor	\$1,903,000	13% (\$243,000)
HENRY ST WPS to QUINN 138 KV	Rebuild line with higher-capacity conductor	\$2,113,000	13% (\$273,000)
YANKE to JARRATT 138 KV line	Rebuild line with higher-capacity conductor	\$7,473,000	13% (\$970,000)
LINDE to RIPPLE 138 KV	Rebuild line with higher-capacity conductor	\$9,697,000	13% (\$1,252,000)
LONG LAKE to MCMINNVILLE 138 KV line	Rebuild line with higher-capacity conductor	\$11,234,000	13% (\$1,451,000)
PRUNEDALE to VALLEYVIEW 138 KV	Rebuild line with higher-capacity conductor	\$15,113,000	14% (\$2,071,000)
PRK AVE to YANKE 138 KV	Rebuild line with higher-capacity conductor	\$5,674,000	13% (\$736,000)
LINDE to MCMINNVILLE 138 KV L	Rebuild line with higher-capacity conductor	\$3,459,000	13% (\$447,000)
EASTON 2 to NAVY MOLE 138 KV	Rebuild line with higher-capacity conductor	\$532,000	13% (\$71,000)
EASTON 2 345/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$16,285,000	13% (\$2,161,000)
MENOMONIE to WATTSVILLE 138 KV	Rebuild line with higher-capacity conductor	\$13,331,000	14% (\$1,824,000)
GRASONVILLE to MACON 138 KV	Rebuild line with higher-capacity conductor	\$2,439,000	13% (\$318,000)
VIENNA OPERATIONS 138 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$6,112,000	14% (\$826,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MENOMONIE to COMMONWEALTH CHESAPEAKE 138 KV line	Rebuild line with higher-capacity conductor	\$7,934,000	14% (\$1,086,000)
ARCOGEN to CHINCOTEAGUE 138 KV	Rebuild line with higher-capacity conductor	\$9,028,000	14% (\$1,235,000)
VIENNA OPERATIONS to SOUTHTOWN 345 KV L	Rebuild line with higher-capacity conductor	\$56,656,000	14% (\$7,705,000)
HUNTINGTON FALLS to HUBBELL 138 KV	Rebuild line with higher-capacity conductor	\$6,386,000	12% (\$757,000)
	Total allocated to project		\$30,127,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.